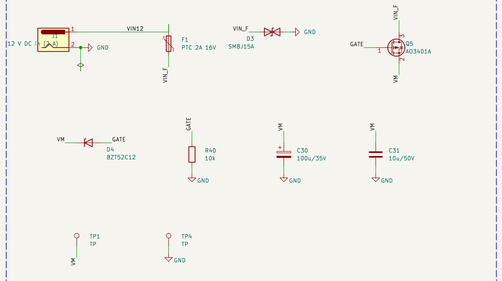
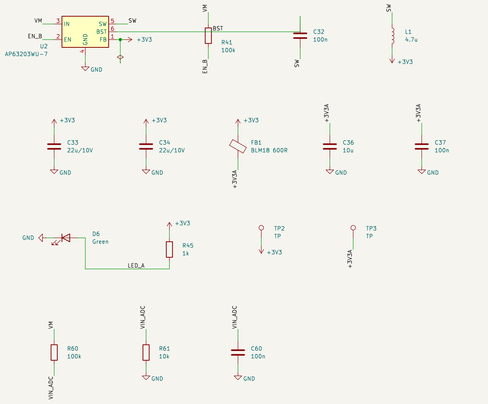
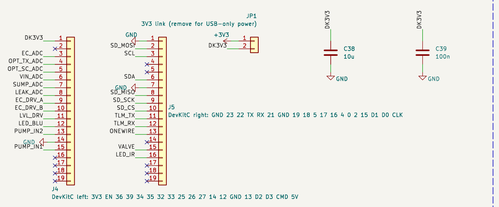
POWER IN: 12 V, 2 A PTC, TVS, reverse-polarity PMOS, bulk



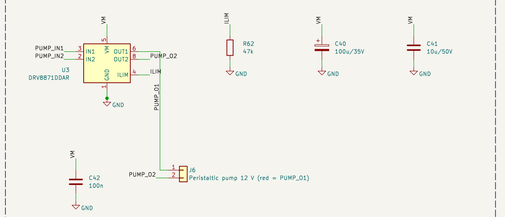
3.3 V BUCK AP63203 + ANALOG RAIL (ferrite) + VIN MONITOR



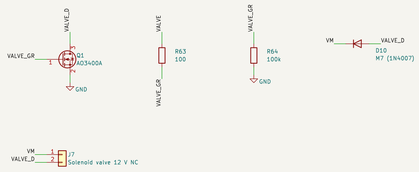
ESP32-DevKitC-32E CARRIER (2 x 1x19 sockets) + 3V3 LINK



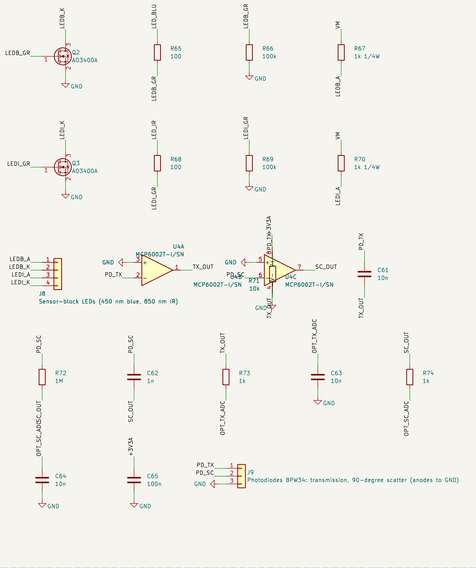
PUMP H-BRIDGE DRV8871 (forward sample / reverse backwash)



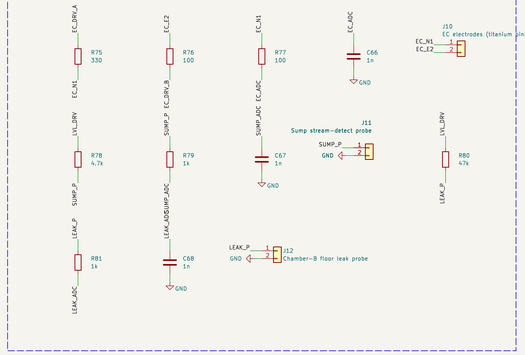
SOLENOID DRIVER AO3400A + 1N4007 FLYBACK



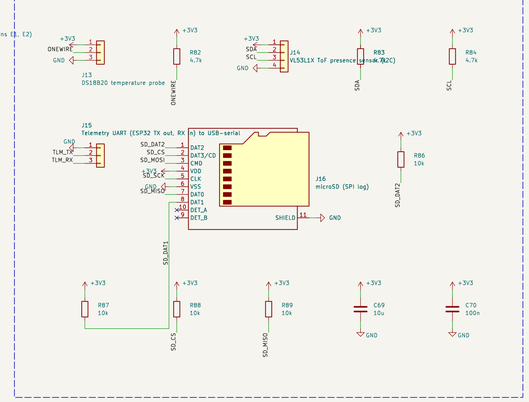
OPTICS: LED DRIVERS + PHOTODIODE TIAs (MCP6002)



EC CELL (bipolar ratiometric), SUMP + LEAK PROBES



TEMP, ToF, TELEMETRY UART, microSD (SPI)



TRISHNA controller, 12 V adapter (2 A) -> PTC -> TVS -> PMOS -> VM. AP63203 makes +3V3; FB1 + 10u/100n make +3V3A for the TIAs. Fit JP1 for normal 12 V operation; REMOVE JP1 before plugging USB into the DevKit. Pump: DRV8871 IN1/IN2 = GPIO13/GPIO12 ((IN1 high = forward/sample, IN2 high = reverse/backwash; both low = coast). Valve: GPIO2 -> AO3400A, M7 flyback across the coil. EC: GPIO25 -> 330 R -> E1 ... E2 -> 100 R -> GPIO26; alternate polarity every 250 us, sample EC_N1 in each phase: (Rc + Rb) / (Rref + Ra) = V_A / V_B, independent of the 3.3 V level; both pins low when idle (no DC on the electrodes). Optics: LEDs from 12 V through 1 k (about 9-10 mA); TIA gains 10 k (transmission) and 1 M (scatter); tune so clear water reads 1.5-2.5 V. Sump/leak probes are pulsed from GPIO27 only while sampled.

First engineering release; see pcb/README.md for status and limits

ClaudeWork	
Sheets: /	
Files: trishna_ctrl.kicad_sch	
Title: TRISHNA urinal hydration monitor – controller (ESP32-DevKitC carrier)	
Size: A1	Date: 2026-11-01
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4	Rev: A
	Id: 1/1